INCH - POUND

A-A-52563
August 15, 1996

COMMERCIAL ITEM DESCRIPTION

PLANIMETERS, POLAR, COMPENSATING, WITH CASE

The General Services Administration has authorized the use of this commercial item description as a replacement for Military Specification MIL-P-52981 for all federal agencies.

1. SCOPE. This specification covers compensating polar planimeters, with cases, that are used for measuring areas and calculating volume from a plane figure by tracing or optically observing its boundary lines.

2. CLASSIFICATION.

Type I - Tracer arm with tracer needle.

Type II - Tracer arm with optical tracer.

3. SALIENT CHARACTERISTICS.

3.1 Description. The planimeter shall be of the compensating polar type. The planimeter shall consist of a tracer arm which carries the tracing point, an electronic measuring mechanism capable of displaying with not less than six decimal places, a switch, and a pole arm around which the instrument revolves. The tracer needle or optical tracer shall be provided with a fingertip guide positioned at the outer end of the tracer arm. A point counter, linear measuring probe, and printing capability shall be available with the planimeter as optional accessories.

3.2 Materials. The contractor shall select the materials, but the materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent possible.

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any other data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

AMSC N/A FSC 6675

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3.2.1 Deterioration prevention and control. The planimeter shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible.

3.2.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Performance characteristics.

3.3.1 Area coverage. When set to its maximum range, the planimeter shall measure the area within a circle having a minimum radius of 16.0-inches (40.6 cm). Measuring range shall not be less than 12.0 x 28.0 inches (30 x 71 cm). Linear measuring range shall be unlimited.

3.3.2 Accuracy. Error of any measurement shall not be greater than 1.0 percent.

3.3.3 Case. The planimeter shall be furnished with a fitted, fungus resistant, carrying case that will protect the instrument from damage.

4. QUALITY ASSURANCE PROVISIONS. The supplier shall comply with the following for certification and acceptability.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2 Market acceptability. The planimeter offered shall be of the latest model of the standard product of the supplier and shall have been used in the field for at least one year by commercial organizations.

4.3 Examination. Each planimeter shall be examined for compliance with the requirements specified in 3.1 through 3.3. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening maximum efficiency shall constitute cause for rejection.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-52563:

AA52563-2

1 - Tracer arm with tracer needle (Type I)
2 - Tracer arm with optical tracer (Type II)

6.2 Ordering Data. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Type of planimeter required
c. When a point counter, linear measuring probe, and printing capability shall be required as optional accessories
d. When the certification of commercial products may be accepted in lieu of testing.

6.3 The procuring agency should specify the level of preservation, packing, and marking required.

6.4 When this Commercial Item Description is used for procurement the contractor certification clause should appear in the solicitation.

6.5 Recovered materials. For the purpose of this requirement, recovered materials are those which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Used, rebuilt, or remanufactured component, pieces and parts shall not be incorporated in the planimeter.

6.6 Subject term (key word) listing.

area measurement
digitizer

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Army - CE
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Review Activity:
Army - CR
DLA - GS

Civil Agency Coordinating Activity:
GSA - FSS

Preparing Activity
Army - CE
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